

The solution to the AI energy consumption problem

Posted by u/Live_Alarm3041 • • 0 points • 9 comments

Photonic computing is the solution to the AI energy consumption issue that has been making headlines in recent times. The ideal way to address the issue is to require all data centers to use photonic chips instead of normal chips. This will drastically reduce the energy demand of AI. Photonic AI chips are already under development by several companies around the world.

Here is the proof

\- <https://news.ufl.edu/2025/09/optical-ai-chip/>

\- [<https://markets.financialcontent.com/stocks/article/tokenring-2025-10-20-beyond-silicon-photonics-and-advanced-materials-forge-the-future-of-semiconductors>]
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The solution to AI energy consumption is not less AI or more carbon neutral energy, it is photonic computing. Photonic computing addresses the energy consumption issue while enabling the growth of AI which is already proving to be beneficial to human society in many ways. The sooner photonic AI chips enter production the sooner we will solve the AI energy consumption issue.

Comments

u/justgord • 6 points • 2025-10-22 01:18 UTC

Thats a pretty shallow analysis .. if photonics were ready I think we'd all be using them.

Likewise theres a lot of hype around small modular nuclear reactors .. maybe they have a role to play, but there sure arent many of them in production use right now.

I think deep drill geothermal has a lot of potential, but it also needs to be commercially developed -

some of these technologies just take a lot of tweaking to get to market scale.

The problem we have is lack of time .. we are at +1.5C global warming now, rising at around +0.3C per decade. At current high plateau of CO2 emissions, we will likely reach **+2.0C around 2040** - only 15 years away.

We really need to take action now with all the technologies we have - solar, wind, battery storage, hydro, reopen closed nuclear plants, move electricity around the country, between countries to even the load.

We need to look at ways to survive the coming peak heat - probably solar radiation management, aka emitting particulates to increase cloud cover over the ocean so sunlight is reflected before being absorbed. this might slow down the heat increase, and buy us time to develop all these other technologies we need.

Of course, we need to move off carbon burning in all its forms, as fast as possible - coal, oil, gas, natural gas etc - and electrify every part of our energy system.

u/mskmagic • 1 points • 2025-10-21 23:04 UTC

Nuclear fusion

u/2nds1st • 1 points • 2025-10-21 21:04 UTC

You want to set up a data center that chews up all the states energy? Build your own power supply for it. Now watch all tje tech bros cry like toddlers that its not fair.

u/creepindacellular • 5 points • 2025-10-21 19:47 UTC

you are wrong, the solution is no AI. no AI, no problem.

for shits and giggles, what trillion dollar problem has AI already solved?

u/HundredHander • 5 points • 2025-10-21 20:46 UTC

I mean it was kind of a problem that science didn't even know if you could spend a trillion dollars, but now we know you can.

u/theonetruefishboy • 19 points • 2025-10-21 17:11 UTC

Dumb tech bullshit no one asked for is the answer to dumb tech bullshit no one asked for.

u/Live_Alarm3041 • -8 points • 2025-10-21 17:13 UTC

So you are against AI in general

If so then alright, I respect your opinion.

u/GnaphaliumUliginosum • 1 points • 2025-10-22 15:50 UTC

I genuinely don't undertand how one can claim to care about ecological impacts and support the development of yet another unnecessary arm of consumer capitalism? Restricting AI to the very small number of potentially socially useful outcomes is the same approach we should have to all polluting and socially harmful industries. Not meant to be an attack, but it does seem like eco-capitalism greenwashing to very slightly mitigate a minority of the harmful impacts of something that doesn't need to exist in the first place.

u/lcy-Piglet-2536 • 1 points • 2026-02-08 09:04 UTC

Idk. I totally agree there's a big missuse and right inflation of the AI bouble. But to want it gone and think it's going to destroy the wrong seems like just a bunch of uneducated old people scared of things they don't understand.

People said the same thing about the Internet and believe it or not even the creation of steam trains. There's always fear about new technologies.

And yet here we are. Internet has become widespread and even though there's still a dangerous side to it, every one uses it. Imagine where society would be right now if we hadn't developed our transport system 100 years ago.

These people saying there should be no AI is like saying our uncestours shouldn't have discovered fire because at the same time it can cook your food it can also burn your house down and kill you.

Every new technology has a learning curve and a development period. We need to learn how to use them properly and development doesn't happen in a week.

I'm an engineer. I work in R&D. We are all extremely aware of the energy problem that AI causes and I can promise everyone that there's plenty of people, both in universities and private companies working on the problem.

u/theonetruefishboy • 10 points • 2025-10-21 19:07 UTC

Oh no I'm pro AI I want every part of the internet to get worse and our civil liberties to be eroded and our jobs to all be destroyed and bought back in forms that pay basically nothing, all so three companies and transfer billions of dollars from dumbass tech investors directly into a giant hole somewhere in Nvidia's basement it's great.

But FR the technology has no positive use case for society at the scale it's currently operating at. I'm sure photographic multi modal reflection sorting or whatever the fuck has some very nice applications, but the solution to the current AI problem is to let the bubble burst, scale waaaaaay the fuck back, and let AIs hundreds to times smaller than ChatGPT sort through MRI data for tumors. Not introduce more tech bullshit no one asked for on top of the other tech bullshit no one asked for.